

12 Mach Union Water Laboratory, Inc.

Address	:	Mach Union Bldg., 335 Alabang-Zapote Road, Talon 3, Las Piñas City
Telephone No.	:	(02) 553-8381; (02) 553-8382; (02) 553-8879; (+63) 9178439306
Fax No.	:	(02) 553-8878
E-mail	:	glenn@machunion.com
Contact Person	:	Glenn U. Pagulayan Laboratory Manager
Field of Testing		Chemical Testing and Microbiological Testing

Chemical Testing

Product	Specific Test	Method Reference
I. Foods		
.01 Cereal and cereal products 1. Breakfast cereals 2. Cereal/cereal grains 3. Cultured seeds 4. Soya flours concentrates and isolates 5. Flour, corn meal, corn grits, semolina 6. Frozen entrees containing rice or corn flour 7. Soy protein 8. Tofu 9. Pasta products and noodles (e.g., rice paper, rice vermicelli, soybean pastas and noodles) 10. Starch	Sodium, Potassium, Calcium, Magnesium	AOAC 985.35 Atomic Absorption Spectroscopy
	Lead, Cadmium, Copper, Iron, Zinc	AOAC 999.11 Atomic Absorption Spectroscopy
	Crude Protein	AOAC 2001.11 Kjeldahl Method
	Crude Fat	AOAC 963.15 Acid Hydrolysis Method AOAC 945.16 Soxhlet Method
	Ash	AOAC 923.03 and 900.0 Dry Ash Method
	Moisture	AOAC 935.29/931.04/945.15 Air-Oven Method AOAC 935.29 Air-Oven and Steam Bath Method AOAC 984.25 Air-Oven and Steam Bath Method
.02 Nuts and nut products 1. Peanut butter and other nut butters	Sodium, Potassium, Calcium, Magnesium	AOAC 985.35 Atomic Absorption Spectroscopy
	Lead, Cadmium, Copper, Iron, Zinc	AOAC 999.11 Atomic Absorption Spectroscopy
	Crude Protein	AOAC 2001.11 Kjeldahl Method
	Crude Fat	AOAC 963.15 Acid Hydrolysis Method

Product	Specific Test	Method Reference
		AOAC 945.16 Soxhlet Method
	Ash	AOAC 923.03 and 900.0 Dry Ashing Method
	Moisture	AOAC 935.29/931.04/945.15 Air-Oven Method AOAC 935.29 Air-Oven and Steam Bath Method AOAC 984.25 Air-Oven and Steam Bath Method
.03 Dairy Products 1. All cheese made from pasteurized milk (cottage cheese, soft and semi-solid cheese) 2. Process cheese spread 3. Ice cream and sherbet plain and flavoured 4. Ice cream with added ingredients 5. Flavored Ice 6. Milk powders (whole, nonfat of filled milk, buttermilk, whey and whey protein concentrate 7. Sweetened condensed milk 8. Liquid milk (evaporated or ready to drink) and Cream (Ultra Heat Temperature/sterilized) 9. Pasteurized Milk 10. Pasteurized cream 11. Yogurt and other fermented milk	Sodium, Potassium, Calcium, Magnesium	AOAC 985.35 Atomic Absorption Spectroscopy
	Lead, Cadmium, Copper, Iron, Zinc	AOAC 999.11 Atomic Absorption Spectroscopy
	Crude Protein	AOAC 2001.11 Kjeldahl Method
	Crude Fat	AOAC 963.15 Acid Hydrolysis Method AOAC 945.16 Soxhlet Method
	Ash	AOAC 923.03 and 900.0 Dry Ashing Method
	Moisture	AOAC 935.29/931.04/945.15 Air-Oven Method AOAC 935.29 Air-Oven and Steam Bath Method AOAC 984.25 Air-Oven and Steam Bath Method
.04 Meat, poultry and derived products 1. Dried animal products 2. Meat pastes and pate' 3. Cold cuts, frozen and chilled hotdogs, corned beef, luncheon meat 4. Packaged cooked cured/salted meat (ham, bacon) 5. Fermented, comminuted meat, no cooked (dry and semi dry fermented sausages) 6. Cooked poultry meat, frozen to be re-heated before eating (e.g., prepared frozen meals 7. Cooked poultry meat, frozen, ready to eat (e.g., turkey rolls)	Sodium, Potassium, Calcium, Magnesium	AOAC 985.35 Atomic Absorption Spectroscopy
	Lead, Cadmium, Copper, Iron, Zinc	AOAC 999.11 Atomic Absorption Spectroscopy
	Crude Protein	AOAC 2001.11 Kjeldahl Method
	Crude Fat	AOAC 963.15 Acid Hydrolysis Method AOAC 945.16 Soxhlet Method
	Ash	AOAC 923.03 and 900.0 Dry Ashing Method
	Moisture	AOAC 935.29/931.04/945.15 Air-Oven Method AOAC 935.29 Air-Oven and Steam Bath Method AOAC 984.25 Air-Oven and

Product	Specific Test	Method Reference
8. Cured/smoked poultry meat 9. Dehydrated poultry products 10. Fresh/frozen raw chicken (during processing) 11. Meat products in hermetically sealed containers		Steam Bath Method
.05 Fish, crustaceans and mollusks and derived products 1. Fresh and frozen fish and cold-smoked 2. Pre-cooked breaded fish 3. Frozen cooked crustaceans 4. Cooked, chilled and frozen crabmeat 5. Fish and shellfish products in hermitically sealed containers 6. Smoked, dried, fermented and salted fish and fish products	Sodium, Potassium, Calcium, Magnesium	AOAC 985.35 Atomic Absorption Spectroscopy
	Lead, Cadmium, Copper, Iron, Zinc	AOAC 999.11 Atomic Absorption Spectroscopy
	Crude Protein	AOAC 2001.11 Kjeldahl Method
	Crude Fat	AOAC 963.15 Acid Hydrolysis Method AOAC 945.16 Soxhlet Method
	Ash	AOAC 923.03 and 900.0 Dry Ashing Method
	Moisture	AOAC 935.29/931.04/945.15 Air-Oven Method AOAC 935.29 Air-Oven and Steam Bath Method AOAC 984.25 Air-Oven and Steam Bath Method
.12 Fruit juices, drinks and concentrates 1. Fruit and vegetable juices 2. Fruit and vegetable nectars 3. Water-based flavoured drinks (Carbonated, Non-carbonated, Concentrates (liquid or solid)) 4. Coffee, coffee substitutes, tea, herbal infusions, and other hot cereal and grain beverages, excluding cocoa	Total Sugar	AOAC 923.09/920.183/945.66 Volumetric Method
	Lead, Cadmium, Copper, Iron, Zinc	AOAC 999.11 Atomic Absorption Spectroscopy
	Sodium, Potassium, Iron, Zinc, Calcium	AOAC 999.11 Atomic Absorption Spectroscopy
	Vitamin C	AOAC Method 967.21 2,6-dichloroindophenol titrimetric method
.25 Food supplement and/ or Dietary supplement	Vitamin A	Laboratory Developed Method (HPLC)
	Vitamin B1, B2, B3 and B6	
	Vitamin C	
	Benzalkonium Chloride	
	Calcium and Magnesium	Laboratory Developed Method (AAS)
III. Drugs and Pharmaceuticals		
.01 Drugs .04 Vitamins	Vitamin A	Laboratory Developed Method (HPLC)
	Vitamin B1, B2, B3 and B6	

Product	Specific Test	Method Reference
	Vitamin C	
	Benzalkonium Chloride	
	Calcium and Magnesium	Laboratory Developed Method (AAS)
VI. Water		
.01 Bottled water	pH	4500-HB. Electrometric/SMEWW 23 rd Ed.
	Calcium, Magnesium, Total Hardness	2340 C. EDTA Titrimetric Method/SMEWW 23 rd Ed.
	Turbidity	2130 B. Nephelometric Method/SMEWW 23 rd Ed
	Total Dissolved Solids	2540 C. Gravimetric (dried @180°C) Method/SMEWW 23 rd Ed
	Mercury	3112 B. Cold Vapor AAS /SMEWW 23 rd Ed
	Arsenic, Cadmium, Copper, Lead, Chromium, Selenium	3113 B. Electrothermal Atomic Absorption Spectroscopy /SMEWW 23 rd Ed
	Aluminum, Barium	3111 D. Nitrous Oxide-Acetylene Flame Method / SMEWW 23 rd Ed.
	Arsenic, Selenium	3114 B. Hydride-Generation Atomic Absorption Spectrometric Method / SMEWW 23 rd Ed.
	Aluminum, Antimony, Arsenic, Barium, Boron, Cadmium, Chromium, Copper, Lead, Manganese, Zinc	3120 B. Inductively Coupled Plasma (ICP) Method / SMEWW 23 rd Ed.
	Chloride, Fluoride, Nitrate, Nitrites, Sulfate	4110 B. Ion Chromatography / SMEWW 23 rd Ed.
	Fluoride	4500-F C. Ion-Selective Electrode Method / SMEWW 23 rd Ed.
	Cadmium, Calcium, Chromium, Copper, Iron, Lead, Manganese, Zinc	3111 B. Flame AAS / SMEWW 23 rd Ed.
	Odor and Taste	Sensory Evaluation
	Nitrate	SM 4500-NO3- B. Ultraviolet Spectrophotometric Screening Method
	Nitrite	SM 4500-NO2- B. Colorimetric Method

Product	Specific Test	Method Reference
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Legend to Reference Methods

AOAC - Association of Official Analytical Chemist

***** *Nothing Follows* *****

Microbiological Testing

Product	Specific Test	Method Reference
I. Foods		
01 Milk and Dairy Products		
01.1 Milk Powders (e.g., whole nonfat or filled milk, buttermilk, whey & whey protein concentrate) (intended for children more than 36 months of age and adults)	Aerobic Plate Count	Conventional Method, BAM Online
01.2 Sweetened Condensed Milk	Aerobic Plate Count	Conventional Method, BAM Online
	Mold and Yeast Count	Conventional Method, BAM Online Method 997.02 Petrifilm OMA AOAC
	Coliforms	Method 991.14 Petrifilm OMA AOAC
01.4 Pasteurized Milk	Aerobic Plate Count	Conventional Method, BAM Online
	Coliforms	Method 991.14 Petrifilm OMA AOAC
01.5 Pasteurized Cream	Aerobic Plate Count	Conventional Method, BAM Online
	Coliforms	Method 991.14 Petrifilm OMA AOAC
01.6 Yogurt and other fermented milk	Coliforms	Method 991.14 Petrifilm OMA AOAC
01.8 Processed Cheese Spread	Aerobic Plate Count	Conventional Method, BAM Online
	Coliforms	Method 991.14 Petrifilm OMA AOAC

Product	Specific Test	Method Reference
02 Fats, Oils, and Fat Emulsions		
02.1 Butter (whipped, pasteurized)	Molds and Yeast Count	Conventional Method, BAM Online Method 997.02 Petrifilm OMA AOAC
	Coliforms	Method 991.14 Petrifilm OMA AOAC
	Aerobic Plate Count	Conventional Method, BAM Online
02.2 Butter made from unpasteurized milk or milk products	Coliforms	Method 991.14 Petrifilm OMA AOAC
	Aerobic Plate Count	Conventional Method, BAM Online
02.3 Margarine	Aerobic Plate Count	Conventional Method, BAM Online
	Molds and Yeast Count	Conventional Method, BAM Online Method 997.02 Petrifilm OMA AOAC
03 Edible Ices including Sherbet and Sorbet		
03.1 Ice Cream & Sherbet (plain and flavored)	Coliforms	Method 991.14 Petrifilm OMA AOAC
	Aerobic Plate Count	Conventional Method, BAM Online
03.2 Ice Cream with added ingredients (nuts, fruits, cocoa, etc.)	Coliforms	Method 991.14 Petrifilm OMA AOAC
	Aerobic Plate Count	Conventional Method, BAM Online
03.3 Flavored Ice (e.g., ice candy)	Aerobic Plate Count	Conventional Method, BAM Online
	Molds and Yeast Count	Conventional Method, BAM Online Method 997.02 Petrifilm OMA AOAC
04 Confectioneries		
04.1 Cocoa Powder	Molds	Conventional Method, BAM Online Method 997.02 Petrifilm OMA AOAC
	Aerobic Plate Count	Conventional Method, BAM Online
04.2 Chocolate Products	Molds	Conventional Method, BAM Online Method 997.02 Petrifilm OMA AOAC
	Aerobic Plate Count	Conventional Method, BAM Online

Product	Specific Test	Method Reference
04.3 Chocolate Confectionaries (chocolate bars, blocks, bonbons)	Molds	Conventional Method, BAM Online Method 997.02 Petrifilm OMA AOAC
	Aerobic Plate Count	Conventional Method, BAM Online
04.4 Sugar Confectionaries (hard & soft candies, caramel, fondants, creams, nougats and pastes)	Molds	Conventional Method, BAM Online Method 997.02 Petrifilm OMA AOAC
	Aerobic Plate Count	Conventional Method, BAM Online
05 Fruits and Vegetables, Nuts and Seeds		
05.2 Fermented Vegetables, ready to eat (e.g., Kimchi)	Mold and Yeast Count	Conventional Method, BAM Online Method 997.02 Petrifilm OMA AOAC
05.5 Desiccated Coconut	Aerobic Plate Count	Conventional Method, BAM Online
	Molds and Yeast Count	Conventional Method, BAM Online Method 997.02 Petrifilm OMA AOAC
05.7 Sun Dried Fruits	Molds	Conventional Method, BAM Online Method 997.02 Petrifilm OMA AOAC
06 Egg and Egg Products		
06.1 Pasteurized Egg Products (liquid, frozen or dried)	Coliforms	Method 991.14 Petrifilm OMA AOAC
	Mold and Yeast Count	Conventional Method, BAM Online Method 997.02 Petrifilm OMA AOAC
	Aerobic Plate Count	Conventional Method, BAM Online
07 Cereals and Cereal Products		
07.1 Breakfast Cereals	Molds Count Yeast & Yeast-like fungi	Conventional Method, BAM Online Method 997.02 Petrifilm OMA AOAC
	Coliforms	Method 991.14 Petrifilm OMA AOAC
	Aerobic Plate Count	Conventional Method, BAM Online

Product	Specific Test	Method Reference
07.2 Cereals/Cereal Grains	Mold and Yeast Count	Conventional Method, BAM Online Method 997.02 Petrifilm OMA AOAC
	Aerobic Plate Count	Conventional Method, BAM Online
	Coliforms	Method 991.14 Petrifilm OMA AOAC
07.3 Cultured seeds and grains (e.g., bean sprouts, alfalfa etc.)	Coliforms	Method 991.14 Petrifilm OMA AOAC
07.4 Soya Flours Concentrates and Isolates	Molds	Conventional Method, BAM Online Method 997.02 Petrifilm OMA AOAC
07.5 Frozen Corn meal, Corn grits, Semolina	Molds Yeast & Yeastlike fungi	Conventional Method, BAM Online Method 997.02 Petrifilm OMA AOAC
	Coliforms	Method 991.14 Petrifilm OMA AOAC
07.7 Soy Protein	Coliforms	Method 991.14 Petrifilm OMA AOAC
	Mold and Yeast Count	Conventional Method, BAM Online Method 997.02 Petrifilm OMA AOAC
	Aerobic Plate Count	Conventional Method, BAM Online
07.9 Pasta Products and Noodles Uncooked (wet & dry)	Coliforms	Method 991.14 Petrifilm OMA AOAC
	Mold and Yeast Count	Conventional Method, BAM Online Method 997.02 Petrifilm OMA AOAC
	Aerobic Plate Count	Conventional Method, BAM Online
07.10 Starch	Coliforms	Method 991.14 Petrifilm OMA AOAC
	Mold and Yeast Count	Conventional Method, BAM Online Method 997.02 Petrifilm OMA AOAC
	Aerobic Plate Count	Conventional Method, BAM Online
08 Bakery Products		
08.3 Frozen and Refrigerated Doughs (Chemically leavened)	Molds Yeast & Yeastlike fungi	Conventional Method, BAM Online

Product	Specific Test	Method Reference
		Method 997.02 Petrifilm OMA AOAC
	Coliforms	Method 991.14 Petrifilm OMA AOAC
	Aerobic Plate Count	Conventional Method, BAM Online
08.4 Frozen and Refrigerated Doughs	Molds Yeast & Yeastlike fungi	Conventional Method, BAM Online Method 997.02 Petrifilm OMA AOAC
	Coliforms	Method 991.14 Petrifilm OMA AOAC
	Aerobic Plate Count	Conventional Method, BAM Online
08.5 Baked Goods (microbiologically sensitive types e.g., containing eggs & dairy products)	Mold and Yeast Count	Conventional Method, BAM Online Method 997.02 Petrifilm OMA AOAC
	Aerobic Plate Count	Conventional Method, BAM Online
	Coliforms	Method 991.14 Petrifilm OMA AOAC
09 Ready to Eat Savouries		
09.1 Snack Foods	Molds Yeast & Yeastlike fungi	Conventional Method, BAM Online Method 997.02 Petrifilm OMA AOAC
	Coliforms	Method 991.14 Petrifilm OMA AOAC
	Aerobic Plate Count	Conventional Method, BAM Online
10.0 Meat and Meat Products		
10.2 Meat paste and Pâté (heat treated)	Coliforms	Method 991.14 Petrifilm OMA AOAC
	Aerobic Plate Count	Conventional Method, BAM Online
10.3 Cold Cuts, Frozen & Chilled Hot Corn Beef, Luncheon Meat	Aerobic Plate Count	Conventional Method, BAM Online
10.9 Fresh/Frozen raw Chicken (during processing)	Aerobic Plate Count	Conventional Method, BAM Online
11.0 Fish and Fish Products		
11.1 Fresh Frozen Fish and Cold- Smoked	Aerobic Plate Count	Conventional Method, BAM Online

Product	Specific Test	Method Reference
11.2 Pre-Cooked Breaded Fish	Aerobic Plate Count	Conventional Method, BAM Online
11.3 Frozen Raw Crustaceans	Aerobic Plate Count	Conventional Method, BAM Online
11.4 Frozen Cooked Crustaceans	Aerobic Plate Count	Conventional Method, BAM Online
11.5 Cooked, Chilled & Frozen Crabmeat	Aerobic Plate Count	Conventional Method, BAM Online
11.6 Fresh and Frozen Bivalve Mollusks	Aerobic Plate Count	Conventional Method, BAM Online
12.0 Spices, Soups, Sauces, Salads, and Protein Products		
12.1 Dry Mixes for Soup and Sauces	Molds and Yeast Count	Conventional Method, BAM Online Method 997.02 Petrifilm OMA AOAC
	Coliforms	Method 991.14 Petrifilm OMA AOAC
	Aerobic Plate Count	Conventional Method, BAM Online
12.3 Spices	Molds	Conventional Method, BAM Online Method 997.02 Petrifilm OMA AOAC
	Aerobic Plate Count	Conventional Method, BAM Online
12.4 Spices (ready to eat)	Coliforms	Method 991.14 Petrifilm OMA AOAC
	Molds	Conventional Method, BAM Online Method 997.02 Petrifilm OMA AOAC
	Aerobic Plate Count	Conventional Method, BAM Online
12.5 Salad Dressing, pH $\leq$ 4.6 (e.g., Mayonnaise, Thousand Island, Ranch, French)	Aerobic Plate Count	Conventional Method, BAM Online
	Molds and Yeast Count	Conventional Method, BAM Online Method 997.02 Petrifilm OMA AOAC
13.0 Beverages		
13.1 Non-Alcoholic Beverages (e.g., ready to drink, soft drinks, iced tea, energy drinks)	Aerobic Plate Count	Conventional Method, BAM Online
	Molds and Yeast Count	Conventional Method, BAM Online

Product	Specific Test	Method Reference
		Method 997.02 Petrifilm OMA AOAC
	Coliforms	Method 991.14 Petrifilm OMA AOAC
13.2 Frozen Juice Concentrate	Aerobic Plate Count	Conventional Method, BAM Online
	Molds and Yeast Count	Conventional Method, BAM Online Method 997.02 Petrifilm OMA AOAC
13.3 Powdered beverages	Aerobic Plate Count	Conventional Method, BAM Online
	Coliforms	Method 991.14 Petrifilm OMA AOAC
14.0 Food for Infants and Young Children		
14.5 Dried and Instant Products requiring reconstitution	Aerobic Plate Count	Conventional Method, BAM Online
14.6 Dried Products requiring reconstitution and boiling before consumption	Coliforms	Method 991.14 Petrifilm OMA AOAC
	Aerobic Plate Count	Conventional Method, BAM Online
14.7 Cereal based foods for infants	Aerobic Plate Count	Conventional Method, BAM Online
IV. Water		
Bottled water	Heterotrophic Plate Count	9215 B. Pour Plate Method / SMEWW 23rd edition
	Total Coliform Count	9221 B Multiple Tube Fermentation Technique/ SMEWW 23rd Ed.
		9223 B. Enzyme Substrate Test / SMEWW 23rd Ed.
	Fecal Coliform Count	9221 E Multiple Tube Fermentation Technique/ SMEWW 23rd Ed.
	<i>E. coli</i> count	9221 G. Multiple Tube Fermentation Technique (Indole Production) / SMEWW 23rd Ed. 9223 B. Enzyme Substrate

Product	Specific Test	Method Reference
		Test (Colilert) / SMEWW 23rd Ed.
	Fecal Streptococcus	9230 B. Multiple Tube Fermentation Technique / SMEWW 23rd Ed.
	<i>Pseudomonas aeruginosa</i>	9213 F. Multiple Tube Fermentation / SMEWW 23rd Ed.

Legend to Reference Methods

BAM - Food and Drug Administration Bacteriological Analytical Manual

OMA - Official Methods of Analysis

AOAC - Association of Official Analytical Chemist

SMEWW - Standard Methods for the Examination of Water and Wastewater

***** *Nothing Follows* *****